APPENDIX 001: STATEMENT OF OBJECTIVES

National Oceanic and Atmospheric Administration (NOAA)

National Environmental Satellite, Data, and Information Service (NESDIS)
Office of Low Earth Orbit Observations (LEO)

Temperature & Moisture Advanced Technology Evolution (TMATE)

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I. Background:

The National Oceanic and Atmospheric Administration (NOAA) is an agency that enriches life through science. NOAA plays several distinct roles within the Department of Commerce (DOC) and focuses on the condition of the oceans and the atmosphere. NOAA is comprised of various line, staff, and program offices that serve as vital contributors to NOAA's mission. NOAA's Mission is to understand and predict changes in climate, weather, oceans, and coasts, to share that knowledge and information with others, and to conserve and manage coastal and marine ecosystems and resources.

The National Environmental Satellite, Data, and Information Service (NESDIS) is a NOAA line office dedicated to providing timely access to global environmental data from satellites and other sources to promote, protect and enhance the Nation's economy, security, environment and quality of life. To fulfill its responsibilities, NESDIS acquires and manages the Nation's operational environmental satellites, operates the NOAA National Data Centers, provides data and information services including earth system monitoring, performs official assessments of the environment, and conducts related research.

Within NESDIS, the Office of Low Earth Orbit Observations (LEO) is responsible for developing the next generation of polar and low earth orbiting environmental satellites to achieve the overarching agency science goals. Atmospheric microwave sounding collected temperature and moisture radiance profiles are considered the most valuable and highest impact data product used in weather forecast modeling by the Numerical Weather Prediction (NWP) community. To increase the refresh rates of these critical observations, the world science community has identified these critical science observations needed to provide a global refresh rate of 6 hours or better. This Request for Project Proposal is to identify solutions to complement the government architecture to improve future weather models.

II. Purpose:

In 2025, NOAA confirmed the agency's minimum threshold for Space-Based LEO Observational Objectives with respect to the National Weather Service (NWS) as requiring imagery and soundings (microwave and infrared) from a minimum of 1330 and 2130 LTAN. Although the Government no longer has resources to populate the historical 1730 LTAN, microwave radiance data from this and other LTANs is highly valuable to improve forecast and model accuracy through an improved refresh rate. NESDIS is therefore looking to gather these observations via alternative means.

To stimulate and advance microwave sounding capabilities, NOAA is interested in pursuing industry-led microwave-based concepts, designs, and/or on-orbit assets to identify and evaluate temperature and moisture radiance profile collection solutions to maintain and improve the threshold requirement for 6-hour global refresh with operational readiness by 2030.

III. Scope or Mission:

This statement of objectives (SOO) describes the scope of work, objectives, and tasks desired by NOAA for producing and providing on-orbit, microwave radiances as well as atmospheric vertical temperature and moisture profiles, with operational capability beginning no later than calendar year 2030.

Delineated in the LEO Observational Objectives, radiance measurements from passive microwave sounding instruments on satellites provide vital input weather prediction and monitoring. NESDIS supports the development of multiple operational algorithms designed for the retrieval of microwave radiances, some in concert with the infrared, some without. While atmospheric vertical temperature and vertical moisture sounding profiles are designated as high priority generated products in terms of user impact, the NESDIS Five Year Product Plan (NESDIS-PLN-1003.1) identifies additional land and hydrology and cryosphere products of interest.

To support generation of these critical NWP products, the solution provider shall be able to acquire and deliver the Level 0 (raw instrument) through Level 1b data (geolocated microwave radiances), as well as generate and deliver the Level 2 (atmospheric vertical temperature and moisture profiles) to the NOAA Ground Segment for data product production, distribution to NWP users, and archival.

The tables below contain threshold and objective microwave sounding radiance requirements (Table 1), vertical temperature profile product requirements (Table 2), and vertical moisture profile product requirements (Table 3) for future observational systems. Table 4 provides spectral guidance for microwave radiance observation.

Activities pursued under this opportunity shall be explicitly delineated and may include (but not be limited to) one or more of the following actions:

- Technology maturation of existing microwave sounding instruments
- Demonstration of instrument performance through ground-based testing
- On-orbit demonstration of temperature and moisture radiance profiling of an existing flight instrument
- Other offeror-proposed solutions

Table 1: Microwave Sounding Radiance Observational Parameters

Performance Level	Threshold	Objective
Nadir Resolution	32 km for Temperature 16 km for Moisture	8 km for Temperature 5 km for Moisture
Calibration Accuracy	2 K	< 1 K
Noise Level (NEDT)	2x ATMS NEDT (< 1.00 @ 57 GHz) (< 1.00 @ 118 GHz) (< 1.00 @ 183 GHz)	< ATMS NEDT (< 0.50 @ 57 GHz) (< 0.50 @ 118 GHz) (< 0.50 @ 183 GHz)
Latency	60 min	15 min
Coverage	Global in 24 hr	Global in 6 hr

Table 2: Atmospheric Vertical Temperature Profile Observational Parameters

System Capability	Threshold
Horizontal Cell Size	50 km
Vertical Reporting Interval	4 km
Mapping Accuracy	5 km
Latency	60 min
Coverage	Global within 24 hr
Temperature Measurement Precision* *Temperature Measurement Precision Expressed as an error in layer average temperature	
Surface to 850 mb over ocean	2.5 K per 1 km Layer
850 to 300 mb over ocean	1.5 K per 1 km Layer
300 mb to 30 mb	1.5 K per 3 km Layer
30 mb to 1 mb	1.5 K per 5 km Layer
1 mb to 0.5 mb	3.5 K per 5 km Layer

Table 3: Atmospheric Vertical Moisture Profile Observational Parameters

System Capability	Threshold
Horizontal Cell Size	16 km
Vertical Reporting Interval	4 km
Mapping Accuracy	5 km
Latency	60 min
Coverage	Global within 24 hr
Moisture Measurement Precision* **Moisture Measurement Precision Expressed as percent of average mixing ratio in 2 km layers.*	
Surface to 600 millibars (mb)	Greater of 40 % or 0.2 grams (g) kilograms (kg)$^{-1}$
600 mb to 300 mb	Greater of 60 % or 0.1 g kg^{-1}
300 mb to 100 mb	Greater of 60 % or 0.1 g kg^{-1}

Table 4: Spectral Guidance for Microwave Radiance Observations

Band Region Name	**Description**	**Primary Use**
Temperature region (T)	V band (50 – 60 GHz) and/or F band near 118 GHz contain O_2 absorptions sensitive to atmospheric profile of temperatures	Vertical profiles of atmospheric temperature; temperature mapping as a function of altitude Includes channels to measure surface temperature
Moisture region (M)	G band near 183 GHz is dominated by water vapor absorptions, used for profiling atmospheric water vapor	Vertical profiles of atmospheric moisture and relative humidity; moisture mapping at multiple altitudes

IV. Period of Performance (POP):

NOAA desires solutions that facilitate on-orbit data collection beginning no later than calendar year 2030 to allow for an on-orbit comparison to Government-provided radiances/products. POP will be dependent on the activities, identified via offeror's whitepaper and negotiated prior to OTA issuance, up to a limit of 24 months.

V. Performance Objectives:

1. Identify innovative, cost-effective approaches to obtain microwave sounding radiances and data products, based on shared contributions from both Government and industry partners.
2. Gain insight into the relative scientific performance of offeror's microwave sounding radiances and data products, consistent with the performance requirements identified in section III, based on the proposed approach.
3. Collaborate with the offeror on OTA execution through, at a minimum:
    a. Periodic meetings with the offeror to discuss significant achievement toward the objectives, technical challenges, and provide added detail and interpretation of NOAA objectives.
    b. Mid-POP progress meeting with the offeror to evaluate status, quality, and advancement toward the objectives, activities planned to complete the performance period and achieve the objectives, and technology/operational capability preview.
    c. End-of-POP presentation with the offeror detailing the solution performance capabilities post-demonstration/maturation, a summary of activities completed during the performance period to achieve the objectives, and an estimated technical approach to on-orbit operation.
    d. Final written report from the offeror including the topics presented at end-of-POP, and a planning estimate of budget and schedule for implementation of the solution from end-of-POP to on-orbit operational.